Lecture 21

Fundamentals of Numerical Analysis

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Better Nodes Example

The first thing we do is SIMPLIFY

- consider the case of $n = 1$ (2-point)
- consider $[a, b] = [-1, 1]$ for simplicity
- we *know* how the trapezoid rule works
- Question: can we possibly do better using only 2 function evaluations?
- Goal: Find w_0, w_1, x_0, x_1 so that

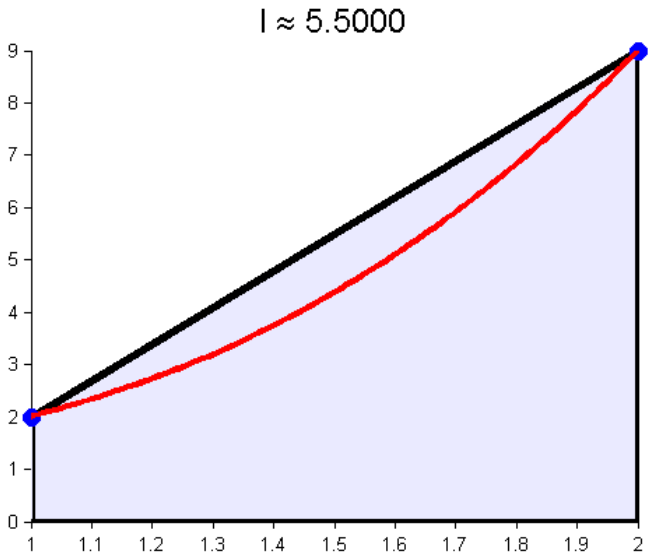
$$\int_{-1}^{1} f(x)\,dx \approx w_0 f(x_0) + w_1 f(x_1)$$

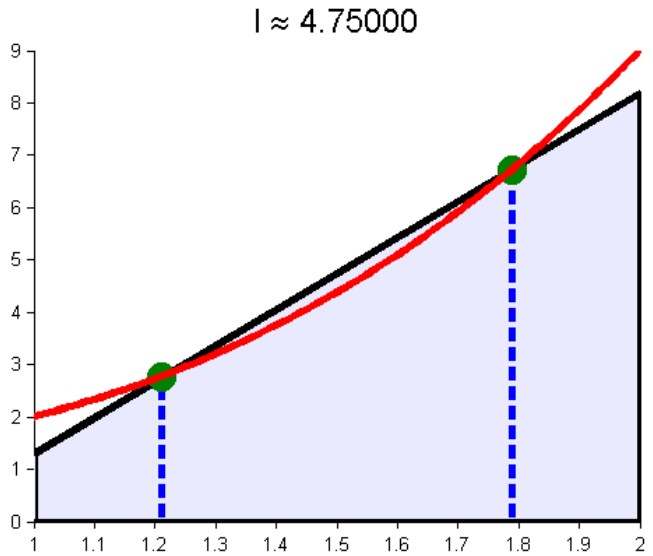
  is as accurate as possible...

Graphical View

Consider

$$\int_1^2 x^3 + 1\, dx = 4.75$$





Derive...

Again, we are considering $[a, b] = [-1, 1]$ for simplicity:

$$\int_{-1}^{1} f(x)\, dx \approx w_0 f(x_0) + w_1 f(x_1)$$

Goal: find w_0, w_1, x_0, x_1 so that the approximation is exact up to cubics. So try any cubic:

$$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

This implies that:

$$\begin{aligned} \int_{-1}^{1} f(x)\, dx &= \int_{-1}^{1} \left(c_0 + c_1 x + c_2 x^2 + c_3 x^3\right)\, dx \\ &= w_0 \left(c_0 + c_1 x_1 + c_2 x_1^2 + c_3 x_1^3\right) + \\ &\quad w_1 \left(c_0 + c_1 x_2 + c_2 x_2^2 + c_3 x_2^3\right) \end{aligned}$$

Derive...

$$\begin{aligned}\int_{-1}^{1} f(x)\,dx &= \int_{-1}^{1} \left(c_0 + c_1 x + c_2 x^2 + c_3 x^3\right)\,dx \\ &= w_0\left(c_0 + c_1 x_1 + c_2 x_1^2 + c_3 x_1^3\right) + \\ &\quad w_1\left(c_0 + c_1 x_2 + c_2 x_2^2 + c_3 x_2^3\right)\end{aligned}$$

Rearrange into constant, linear, quadratic, and cubic terms:

$$\begin{aligned}c_0\left(w_0 + w_1 - \int_{-1}^{1} dx\right) &+ c_1\left(w_0 x_0 + w_1 x_1 - \int_{-1}^{1} x\,dx\right) + \\ c_2\left(w_0 x_0^2 + w_1 x_1^2 - \int_{-1}^{1} x^2\,dx\right) &+ c_3\left(w_0 x_0^3 + w_1 x_1^3 - \int_{-1}^{1} x^3\,dx\right) = 0\end{aligned}$$

Since c_0, c_1, c_2 and c_3 are arbitrary, then their coefficients must all be zero.

Derive...

This implies:

$$w_0 + w_1 = \int_{-1}^{1} dx = 2 \qquad w_0 x_0 + w_1 x_1 = \int_{-1}^{1} x\, dx = 0$$

$$w_0 x_0^2 + w_1 x_1^2 = \int_{-1}^{1} x^2\, dx = \frac{2}{3} \qquad w_0 x_0^3 + w_1 x_1^3 = \int_{-1}^{1} x^3\, dx = 0$$

Some algebra leads to:

$$w_0 = 1 \quad w_1 = 1 \quad x_0 = -\frac{\sqrt{3}}{3} \quad x_1 = \frac{\sqrt{3}}{3}$$

Therefore:

$$\int_{-1}^{1} f(x)\, dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right)$$

Over another interval?

$$\int_{-1}^{1} f(x)\,dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right)$$

$$\int_{a}^{b} f(x)\,dx \approx ?$$

- integrating over $[a, b]$ instead of $[-1, 1]$ needs a transformation: a change of variables
- want $t = c_1 x + c_0$ with $t = -1$ at $x = a$ and $t = 1$ at $x = b$
- let $t = \frac{2}{b-a}x - \frac{b+a}{b-a}$
- (verify)
- let $x = \frac{b-a}{2}t + \frac{b+a}{2}$
- then $dx = \frac{b-a}{2}dt$

Over another interval?

$$\int_a^b f(x)\,dx \approx ?$$

- let $x = \frac{b-a}{2}t + \frac{b+a}{2}$
- then $dx = \frac{b-a}{2}dt$

$$\int_a^b f(x)\,dx = \int_{-1}^{1} f\left(\frac{(b-a)t+b+a}{2}\right)\frac{b-a}{2}\,dt$$

- now use the quadrature formula over $[-1,1]$
- note: using two points, $n = 1$, gave us exact integration for polynomials of degree less 2*1+1 = 3 and less.

Extending Gauss Quadrature

- we need more to make this work for more than two points
- A sensible quadrature rule for the interval $[-1, 1]$ based on 1 node would use the node $x = 0$. This is a root of $\phi(x) = x$
- Notice: $\pm\frac{1}{\sqrt{3}}$ are the roots of $\phi(x) = 3x^2 - 1$
- general $\phi(x)$?

Gauss Quadrature Theorem

Karl Friedrich Gauss proved the following result:
Let $q(x)$ be a nontrivial polynomial of degree $n+1$ such that

$$\int_a^b x^k q(x)dx = 0 \qquad (0 \leqslant k \leqslant n)$$

and let $x_0, x_1, \ldots, x_n$ be the zeros of $q(x)$. Then

$$\int_a^b f(x)dx \approx \sum_{i=0}^{n} A_i f(x_i), A_i = \int_a^b \ell_i(x)dx$$

will be exact for all polynomials of degree at most $2n+1$. (Wow!)

Sketch of Proof

Let $f(x)$ be a polynomial of degree $2n+1$. Then we can write $f(x) = p(x)q(x) + r(x)$, where $p(x)$ and $r(x)$ are of degree at most n (This is basically dividing f by q with remainder r).

Then by the hypothesis, $\int_a^b p(x)q(x)dx = 0$. Further, $f(x_i) = p(x_i)q(x_i) + r(x_i) = r(x_i)$. Thus,

$$\int_a^b f(x)dx = \int_a^b r(x)dx \approx \sum_{i=0}^{n} f(x_i) \int_a^b \ell_i(x)dx$$

But this is exact because $r(x)$ is (at most) a degree n polynomial. Thus, we need to find the polynomials $q(x)$.

Orthogonal Polynomials

Orthogonality of Functions

Two functions $g(x)$ and $h(x)$ are *orthogonal* on $[a, b]$ if

$$\int_a^b g(x)h(x)\,dx = 0$$

- so the nodes we're using are roots of orthogonal polynomials
- these are the *Legendre* Polynomials

Legendre Polynomials

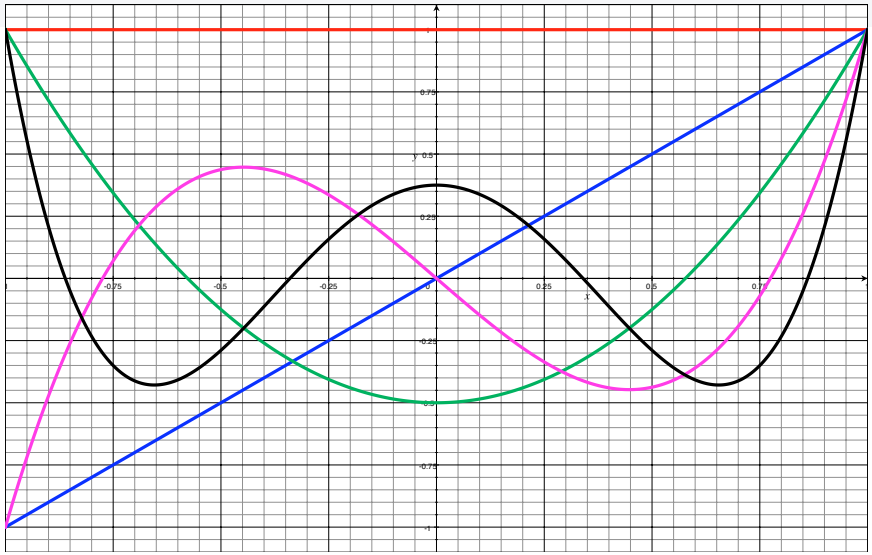
given on the exam

$$\phi_0 = 1$$
$$\phi_1 = x$$
$$\phi_2 = \frac{3x^2 - 1}{2}$$
$$\phi_3 = \frac{5x^3 - 3x}{2}$$
$$\vdots$$

In general:

$$\phi_n(x) = \frac{2n-1}{n} x\phi_{n-1}(x) - \frac{n-1}{n}\phi_{n-2}(x)$$

Notes on Legendre Roots



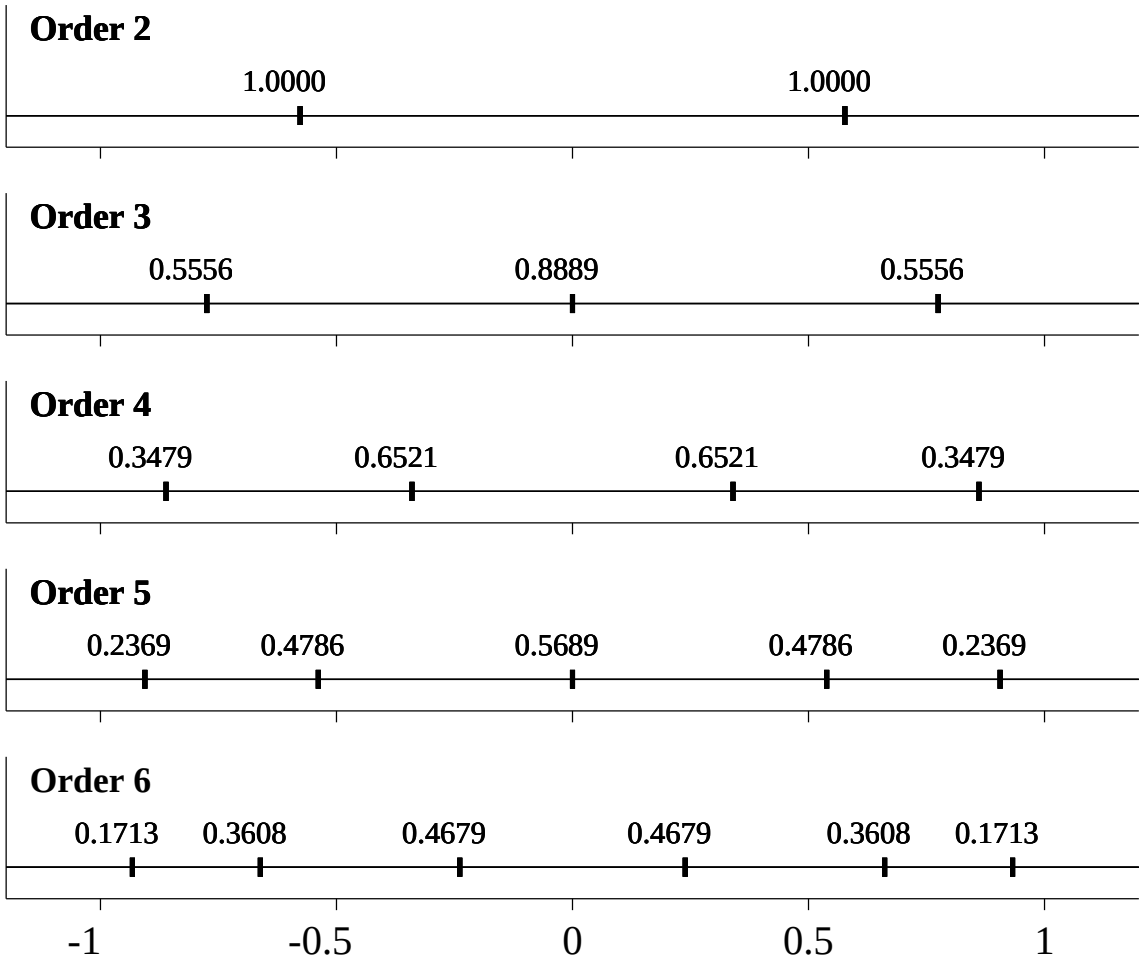
- The Legendre Polynomials are orthogonal (nice!)
- The Legendre Polynomials increase in polynomials order (like monomials)
- The Legendre Polynomials don't suffer from poor conditioning (unlike monomials...more in the linear algebra section)
- The Legendre Polynomials don't have a closed form expression (recursion relation is needed)
- The roots of the Legendre Polynomials are the nodes for Gaussian Quadrature (GL nodes)

Quadrature Nodes (see)

- Often listed in tables
- Weights determined by extension of above
- Roots are symmetric in $[-1, 1]$
- Example:

```
if(n==0)
  x = 0;   w = 2;
if(n==1)
  x(1) = -1/sqrt(3);     x(2) = -x(1);
  w(1) =  1;             w(2) =  w(1);
if(n==2)
  x(1) = -sqrt(3/5);    x(2) =  0;      x(3) = -x(1);
  w(1) =  5/9;          w(2) =  8/9;    w(3) =  w(1);
if(n==3)
  x(1) = -0.861136311594053;     x(4) = -x(1);
  x(2) = -0.339981043584856;     x(3) = -x(2);
  w(1) =  0.347854845137454;     w(4) =  w(1);
  w(2) =  0.652145154862546;     w(3) =  w(2);
if(n==4)
  x(1) = -0.906179845938664;     x(5) = -x(1);
  x(2) = -0.538469310105683;     x(4) = -x(2);
  x(3) =  0;
  w(1) =  0.236926885056189;     w(5) =  w(1);
  w(2) =  0.478628670499366;     w(4) =  w(2);
  w(3) =  0.568888888888889;
if(n==5)
  x(1) = -0.932469514203152;     x(6) = -x(1);
  x(2) = -0.661209386466265;     x(5) = -x(2);
  x(3) = -0.238619186083197;     x(4) = -x(3);
  w(1) =  0.171324492379170;     w(6) =  w(1);
  w(2) =  0.360761573048139;     w(5) =  w(2);
  w(3) =  0.467913934572691;     w(4) =  w(3);
```

View of Nodes



Theory

The connection between the roots of the Legendre polynomials and exact integration of polynomials is established by the following theorem.

Theorem

Suppose that $x_0, x_1, \ldots, x_n$ are roots of the nth Legendre polynomial $P_n(x)$ and that for each $i = 0, 1, \ldots, n$ the numbers w_i are defined by

$$w_i = \int_{-1}^{1} \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} dx = \int_{-1}^{1} \ell_i(x)\, dx$$

Then

$$\int_{-1}^{1} f(x)dx = \sum_{i=0}^{n} w_i f(x_i),$$

where $f(x)$ is any polynomial of degree less or equal to $2n + 1$.

Do not!

!!!

When evaluating a quadrature rule

$$\int_{-1}^{1} f(x)dx = \sum_{i=0}^{n} w_i f(x_i).$$

do not generate the nodes and weights each time. Use a lookup table...

Example

Approximate $\int_1^{1.5} x^2 \ln x \, dx$ using Gaussian quadrature with $n = 1$.

Solution As derived earlier we want to use $\int_{-1}^{1} f(x) \, dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right)$

From earlier we know that we are interested in

$$\int_1^{1.5} f(x) \, dx = \int_{-1}^{1} f\left(\frac{(1.5-1)t + (1.5+1)}{2}\right) \frac{1.5-1}{2} \, dt$$

Therefore, we are looking for the integral of

$$\frac{1}{4}\int_{-1}^{1} f\left(\frac{x+5}{4}\right) dx = \frac{1}{4}\int_{-1}^{1} \left(\frac{x+5}{4}\right)^2 \ln\left(\frac{x+5}{4}\right) dx$$

Using Gaussian quadrature, our numerical integration becomes:

$$\frac{1}{4}\left[\left(\frac{-\frac{\sqrt{3}}{3}+5}{4}\right)^2 \ln\left(\frac{-\frac{\sqrt{3}}{3}+5}{4}\right) + \left(\frac{\frac{\sqrt{3}}{3}+5}{4}\right)^2 \ln\left(\frac{\frac{\sqrt{3}}{3}+5}{4}\right)\right] = 0.1922687$$

Example

Approximate $\int_0^1 x^2 e^{-x}\, dx$ using Gaussian quadrature with $n = 1$.

Solution We again want to convert our limits of integration to -1 to 1. Using the same process as the earlier example, we get:

$$\int_0^1 x^2 e^{-x} dx = \frac{1}{2} \int_{-1}^1 \left(\frac{t+1}{2} \right)^2 e^{(t+1)/2} dt.$$

Using the Gaussian roots we get:

$$\int_0^1 x^2 e^{-x} dx \approx \frac{1}{2} \left[\left(\frac{-\frac{\sqrt{3}}{3} + 1}{2} \right)^2 e^{(-\frac{\sqrt{3}}{3}+1)/2} + \left(\frac{\frac{\sqrt{3}}{3} + 1}{2} \right)^2 e^{(\frac{\sqrt{3}}{3}+1)/2} \right] = 0.1594104$$

Numerical Question

How does n point Gauss quadrature compare with n point Newton-Cotes...

Examples

with matlab...

Example

int_gauss.m: base routine for Gauss quadrature

Example

int_gauss_test.m: integrate $\int_0^5 xe^{-x}\,dx$ with

1. 1 subinterval, increasing number of nodes
2. 3 nodes, increases number of intervals (panels)

Result: fewer total evaluations in GL quadrature with 1 subinterval and many nodes versus 3 nodes and many subpanels. Also more accurate.

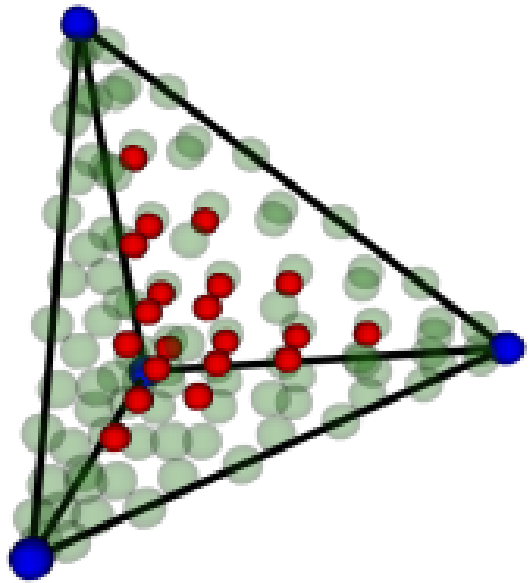
Example

int_compare_gauss_trapezoid_simpson.m: integrate $\int_0^5 xe^{-x}\,dx$ with

1. trapezoid
2. Simpson
3. Gauss

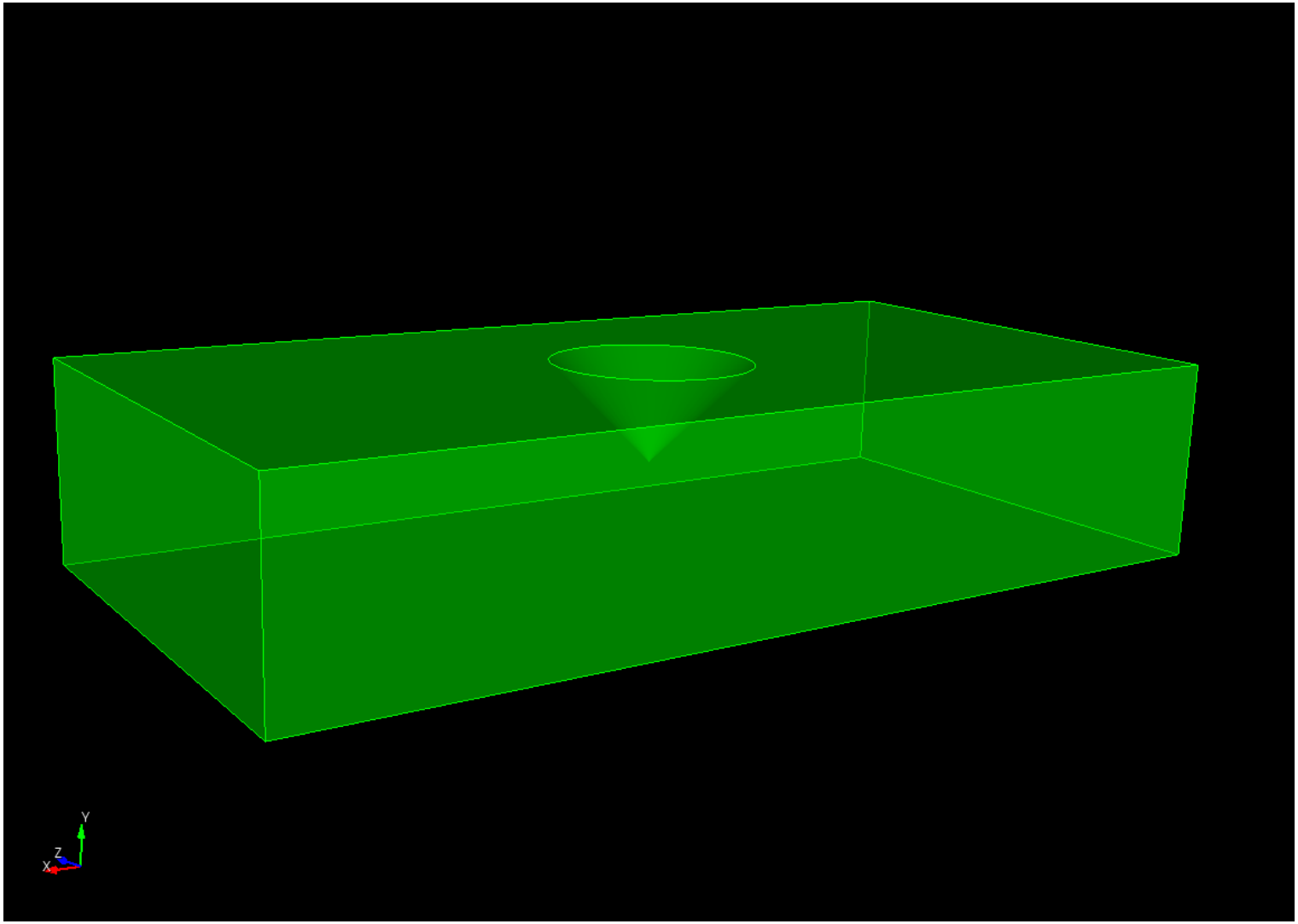
Keep the number of number of nodes and panels consistent. Result: fewer evals for Gauss

Something interesting...

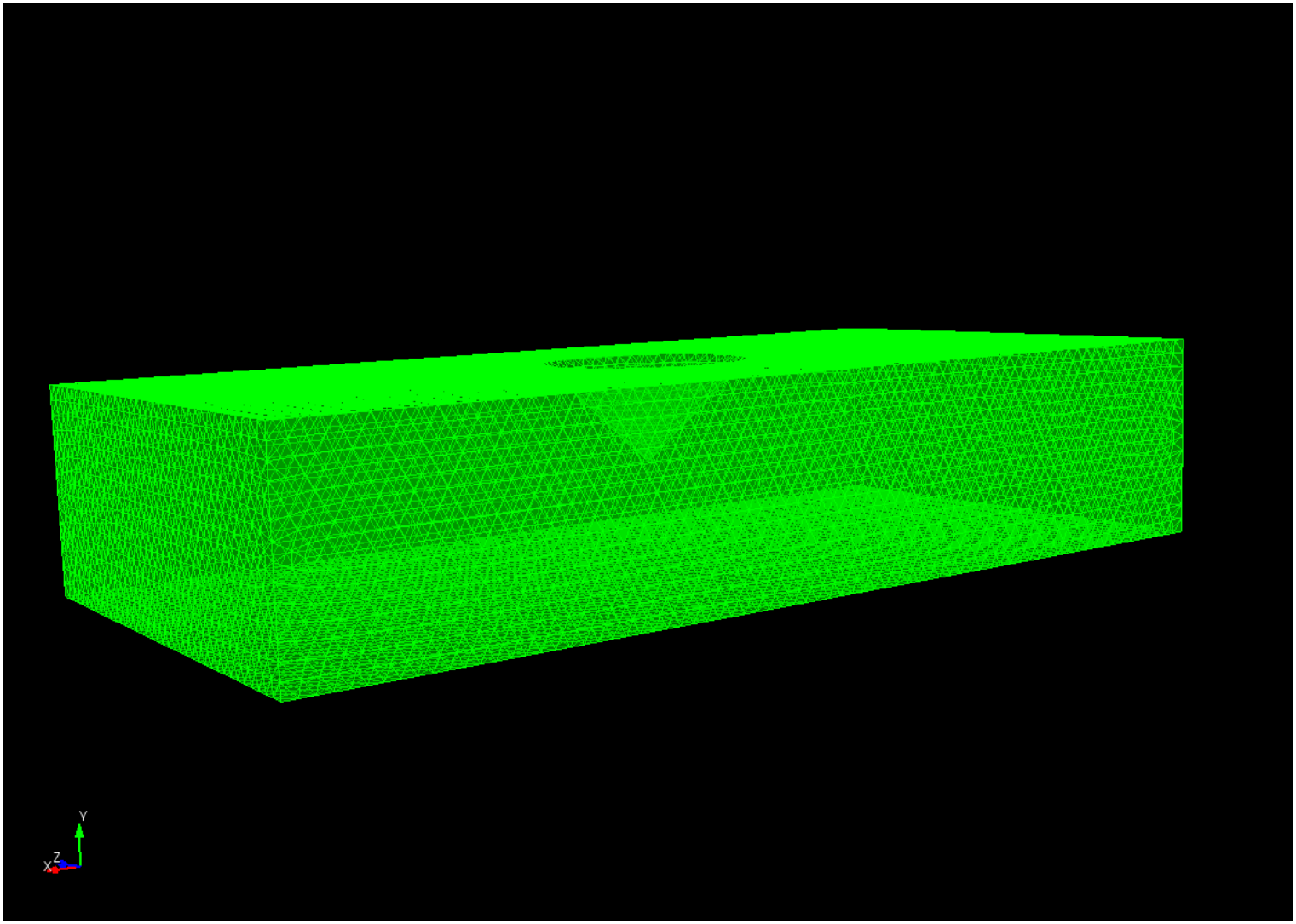


- Gauss nodes in other shapes are not easy
- One approach: let electrostatics determine the (uneven) distribution

Quadrature

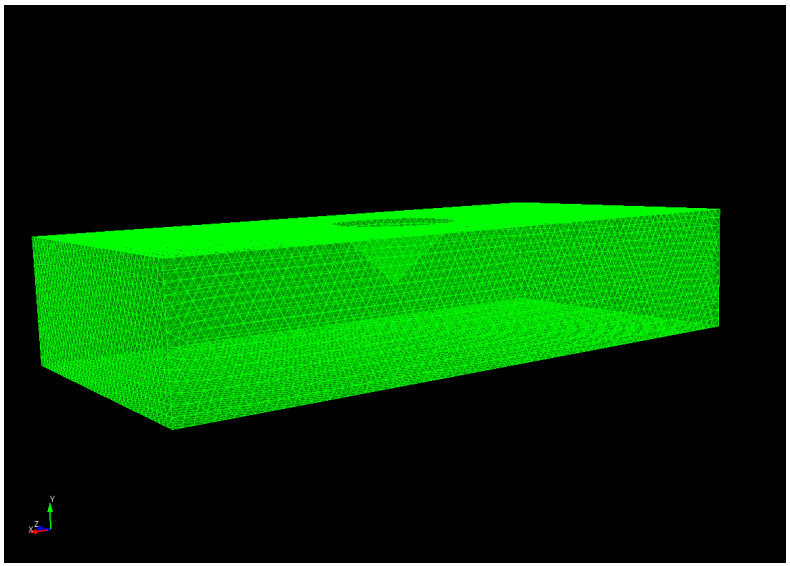


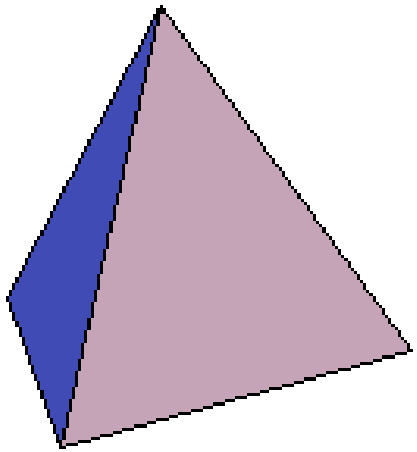
Quadrature



40K tetrahedrons.

Quadrature





- 40K tetrahedrons.
- need to integrate a function $f(x)$ over each tet Ω_i: $\int_{\Omega_i} f(x)\,dx$
- needs quadrature
- map the integration to a reference tetrahedron
- perform Gauss Quadrature using $(n+1)(n+2)(n+3)/6$ quadrature points (where n is the 1-D number of points).
- $f(x)$ is often a product of degree m Lagrange interpolating polynomials
- ...needs large n
- results in # of tetrahedrons $\times\ n^3/6$ function evaluations